Alessandro Codega

Curriculum Vitae

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Research Profile

Master's graduate in Theoretical Physics specializing in high energy physics, Quantum Field Theories and the Standard Model, Higgs phenomenology and Beyond Standard Model searches. Experience in Monte Carlo simulations, machine learning techniques, and phenomenology. Scientific outreach enthusiast, amateur viola da gamba player.

Education

2023 - 2026 **Master's Degree in Physics**, University of Pavia, Italy.
Final Grade: 106/110, defended 12th June 2026
GPA: 28.83/30
Thesis: *A Phenomenological Study for $H \to Za$ Searches*,
supervisors: Prof. Daniela Rebuzzi, Dr. Sonia Carrà, Dr. Giovanni Stagnitto

2019 - 2023 **Bachelor's Degree in Physics**, University of Pavia, Italy.
Final Grade: 100/110.
Thesis: *Constraining the Neutrino Sector*,
supervisor: Prof. Daniela Rebuzzi.

2019 **Diploma di Istruzione Secondaria Superiore** (Scientific High School Diploma), Liceo Scientifico Carlo Donegani, Sondrio, Italy.
Final Grade: 100/100.

Research Experience

Master's Thesis **A Phenomenological Study for $H \to Za$ Searches, Using Monte Carlo Generators and a Two-Stage Neural Network Pipeline**
Key contributions:

- Generated signal and background Monte Carlo samples using POWHEG, PYTHIA8, and HERWIG7, with fast detector simulation via DELPHES;
- Performed a systematic comparison of PYTHIA8 and HERWIG7 generators to quantify generator-tied modeling uncertainties in Za production;
- Applied `FastJet` and `SoftDrop` grooming algorithms for jet reconstruction and substructure analysis;
- Designed and implemented a two-stage neural network pipeline (regression + classification) in Python to improve signal-background separation and reduce theoretical systematic uncertainties;
- Developed jet-tagging variables and machine learning classifiers, achieving stable cross-generator AUC performance in the groomed configuration;
- Analyzed and compared Monte Carlo samples in ROOT to assess systematic uncertainties across generator configurations.

Publications and Presentations

17th Feb 2025 — **Poster Presentation**, Department of Physics, Pavia, Italy.
Topic: *Study of the Exotic Higgs Decay $H \to Za$.*
Poster presentation at a physics department congress, presenting ongoing research on Higgs phenomenology within the 2HDM framework.

Physics Education Journal — L. Zatti, C. Aimè, D. Aurelio, E. Budassi, A. Codega, P. Montagna, S. Restelli,
A Taste of Physics: The Scientific Method in the Kitchen,
Manuscript in preparation.

GIREP 2026 — L. Zatti, C. Aimè, D. Aurelio, E. Budassi, A. Codega, P. Montagna, S. Restelli,
A Taste of Physics: The Scientific Method in the Kitchen,
Talk accepted, GIREP 2026 Conference, Rethymno, Crete, Greece.

7th Sep 2026 — A. Codega, S. Carrà, D. Rebuzzi, G. Stagnitto,
A Phenomenological Study for $H \to Za$ Searches,
Oral presentation, SIF Florence 2026 Conference, Florence, Italy.

Professional Experience

Feb 2026 — **Tutor – ATLAS Particle Physics Masterclass**, Pavia, Italy.

The Particle Physics Masterclasses include exercises that allow students to select events and learn key concepts of analysis. Assisted high school students by helping them identify particle events and answering related questions.

Oct 2024 - Jun 2025 — **Tutor – INFN Lab2Go Project**, INFN, Pavia, Italy.

Selection through competitive scholarship for the INFN Lab2Go outreach project. Contribution to the requalification of high school physics laboratories of the province of Pavia, Italy. Elaboration and explanation of experimental physics activities for students. Collaboration with teachers to adapt laboratory activities to high school curricular goals. The project was held in joint activity with the Italian "Alternanza Scuola-Lavoro" project.

Oct - Nov 2025 — **Scientific Explainer crew member**, Festival della Scienza di Genova, Italy.
Title: *"Fisica da leccarsi i baffi"*.

Public outreach activity focused on communicating physics through everyday kitchen phenomena and appliances. Measurement of the speed of light with a microwave and a chocolate bar, measurement of exponential beer froth decay, measurement of the "hot chocolate effect". Interaction with high school students and a good training ground for scientific communication.

Oct - Nov 2024 — **Scientific Explainer crew member**, Festival della Scienza di Genova, Italy.
Title: *"A caccia di fotoni gamma con l'INAF"*.

Assistance with live demonstrations and interactive exhibits about the Cherenkov Telescope Array Observatory (CTAO), explaining complex physics topics and phenomena to the general public. INAF's outreach magazine article about this experience available here.

Outreach and Volunteer Activities

Sep 2023 **European Researcher's Night**, Pavia, Italy.

Collaboration with the Physics4Teenagers Pavia group in presenting scientific outreach activities for the general public. Assistance in the preparation and direction of the *Escape Room CNAO*, an interactive experience on biomedical physics, designed by the CNAO (*Centro Nazionale Adroterapia Oncologica*) association.

Sep 2022 **European Researcher's Night**, Pavia, Italy.

Collaboration with the Physics4Teenagers Pavia group in presenting scientific outreach activities for the general public during the European Researcher's Night event.

Technical Skills

Programming Python (NumPy, pandas, scikit-learn, matplotlib), MATLAB, LaTeX, C++, Fortran.

Scientific Data analysis and statistical methods; Monte Carlo event generators: POWHEG, PYTHIA8, HERWIG7; fast detector simulation: DELPHES; data analysis framework: ROOT; machine learning for particle physics; generator tuning and systematic uncertainty evaluation.

Languages

Italian Native.

English Proficient (C2).

French Basic (A1).

Certifications

2018 Cambridge English First (C1).

Pavia, June 17th 2026.